COMPUTING Ω_1: THE $O(1/\beta)$ NOISE-AVERAGED SHIFT OF THE AIRY CONNECTION ROOT (COMPLETING T2B,C FOR TW_β)

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Abstract. The O5 note reduced Theorem T2 to a single finite second-Wiener-chaos integral of the Airy Green's function. We evaluate it: $\Omega_1 = -1.12$. This is the $O(1/\beta)$ noise-averaged shift of the Airy connection root (ground-state eigenvalue), the direct transpose of the cusp Ω_2, computed via the sum-over-states form of $-4\int \psi_0^2\, G_{\mathrm{red}}$, which converges (terms $\sim n^{-4/3}$) precisely because the Green's function spreads the noise contraction — unlike the $\delta(0)$-divergent period.

1. The integral and its value

For the stochastic Airy operator $-\partial_x^2 + x + 2\varepsilon b'$ on $\mathbb{R}_+$ (Dirichlet, $\varepsilon = \beta^{-1/2}$), second-order Rayleigh–Schrödinger perturbation theory with $W = 2\varepsilon b'$ gives (the first-order mean vanishes since $\mathbb{E}[b'] = 0$)

$$\mathbb{E}[\Lambda_0] = \Lambda_0^{\mathrm{det}} + \varepsilon^2\,\Omega_1 + O(\varepsilon^4), \qquad \Omega_1 = 4\sum_{n\ge 1} \frac{\langle \psi_0^2 \mid \psi_n^2\rangle}{\Lambda_0 - \Lambda_n} = -4\int_0^\infty \psi_0(x)^2\, G_{\mathrm{red}}(x,x)\,\mathrm{d}x, \tag{1}$$

where $\psi_n(x) = \mathrm{Ai}(x-\Lambda_n)/\|\cdot\|$, $\Lambda_n = |(n{+}1)\text{-th zero of Ai}|$ ($\Lambda_0 = 2.3381$), $\|\mathrm{Ai}(\cdot - \Lambda_n)\|^2 = \mathrm{Ai}'(-\Lambda_n)^2$, and G_{red} is the reduced resolvent at Λ_0.

Proposition 1 (Value)**.** *$\Omega_1 = -1.12 \pm 0.01$, hence $\mathbb{E}[\Lambda_0](\beta) = 2.3381 - 1.12/\beta + O(\beta^{-2})$.* [computed]Two independent methods agree: *the sum-over-states* (1) *gives* -1.121 *(partial sum* -1.010 *to* $n{=}299$ *plus power-law tail* -0.110*); a direct-diagonalization Monte Carlo — diagonalizing* $H_0 + \mathrm{diag}(2\varepsilon b')$ *over noise realizations with a variance-reduced second-order estimator, using neither the eigenstate sum nor a Green's function — gives* -1.128 ± 0.009 *(*ε*-independent over* $\varepsilon \in \{0.03, 0.05, 0.08\}$*, grid-converging to* ≈ -1.13*). Agreement to* $< 1\%$*.*

The sum (1) converges: the overlaps decay as $\langle\psi_0^2|\psi_n^2\rangle \sim (2\Lambda_n)^{-1}$ (WKB density of a high Airy state at fixed x), and $\Lambda_0 - \Lambda_n \sim -\Lambda_n \sim -n^{2/3}$, so the summand is $\sim n^{-4/3}$ — *convergent*, verified numerically (fitted decay exponent $p = 1.350 \approx \frac{4}{3}$; partial sum -1.010 to $n = 299$, plus a power-law tail -0.110, stable to ± 0.001 across fit windows). This is the finiteness the O5 note asserted: although $\sum_n \langle\psi_0^2|\psi_n^2\rangle = \delta(0)\int \psi_0^2$ diverges (the coincident-point self-contraction that makes the *period*/location shift $\delta(0)$-divergent), the energy denominator $1/(\Lambda_0 - \Lambda_n)$ renders the *connection-root* shift finite. [computed]

Remark 2 (What Ω_1 is, and the honest caveat)*.* Ω_1 is the $O(1/\beta)$ shift of the Airy *connection root* $\{D(\Lambda) = 0\}$ — the exact transpose of the cusp $\Omega_2 = \mathbb{E}[\text{connection-root shift}]$, computed by the same second-Wiener-chaos/Green's-function structure. One structural difference: the cusp resonance root $\lambda_0 = 0.8896(1-i)$ is *complex* (oscillatory Weber connection), so $\Omega_2 = -0.451+0.352i$ carries Stokes/phase information; the Airy connection root is the *real* ground-state eigenvalue, so Ω_1 is real — the noise-averaged mean shift. It completes T2(b,c) at the level of the connection-root datum; the genuinely resurgent (complex) Stokes constant lives on the oscillatory sheet of Painlevé II ($s \to -\infty$) and is a further computation on that branch. [caveat]

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Status. Theorem T2(b,c) — "$\Omega(\beta) = \Omega_{\mathrm{HM}} + \beta^{-1}\Omega_1 + \cdots$ with Ω_1 a finite renormalized Airy-Green's chaos integral" — is now *evaluated*: $\Omega_1 = -1.12$. Combined with the O5 location-fixing lemma (T2a), Theorem T2 is established at the connection-root level, and the value $\Omega_1 = -1.12$ is cross-checked by two independent methods (Prop. 1). The one remaining refinement is the complex Stokes constant on the oscillatory PII branch (Rem. 2). Tools: `coupled-atlas/_omega1_airy.py` (sum-over-states), `_omega1_crosscheck.py` (diagonalization MC).